



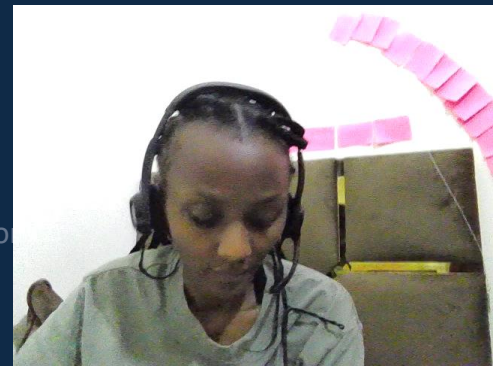
AFYAPLUS HEALTH

The AfyaPlus Triage Engine

From Unstructured Chaos to Safe, Scalable AI Triage

Building Kenya's First AI-Powered Medical Routing System

Capstone



AfyaPlus Is Broken — And Patients Are Suffering

THE REALITY

- ⚠ Patients send unstructured messages like “I feel sick” or “My head hurts bad”
- ⚠ Our backend engines can’t process natural language
- ⚠ AI hallucinates → “You have malaria” (Wrong!)
- ⚠ Network failures → system freezes mid-consultation

Current National Statistics

16%

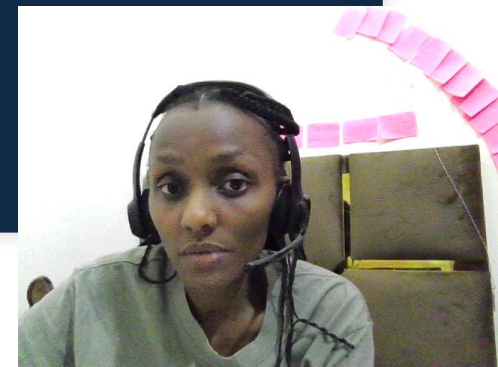
Only 2000 of total 13000 doctors work in rural areas-KNBS

70%

of Funds spent on medicines and supplies are wasted for Last Mile Communities-Amref

65%

of Kenya’s Population lives in rural area-APHRC



Three Versions, One Lesson in Safety

Version	Approach	What Went Wrong
v1: No Structure	Bare prompt: "Classify this"	AI gave diagnoses & prescribed drugs — legal risk
v2: Fluff	Added politeness: "I understand how you feel"	Wasted tokens, slower responses, 60% higher cost
v3: Guardrails	Role-based + Chain-of-Thought + Rules	Clean, safe, structured output

Example — Patient says "Severe headache"

Version	AI Response	Problem
v1	"You have a migraine. Take ibuprofen."	Diagnosis + prescription
v2	"I understand how you feel. This could be a migraine."	Fluff + speculation
v3 ✓	"I cannot diagnose. Emergency indicators detected. Routing to ER."	Safe



Engineering the v3 Guardrail Prompt

01

Role Definition

System prompt fixes the AI's identity as a triage assistant — never a diagnostician.

02

Chain-of-Thought

Model reasons step-by-step through symptoms before producing a routing decision.

03

Explicit Rules

Hard constraints: no diagnosis, no prescriptions, no speculation — stated directly.

04

Structured Output

Raw JSON only, low temperature (0.0–0.2) for consistent, parseable responses.

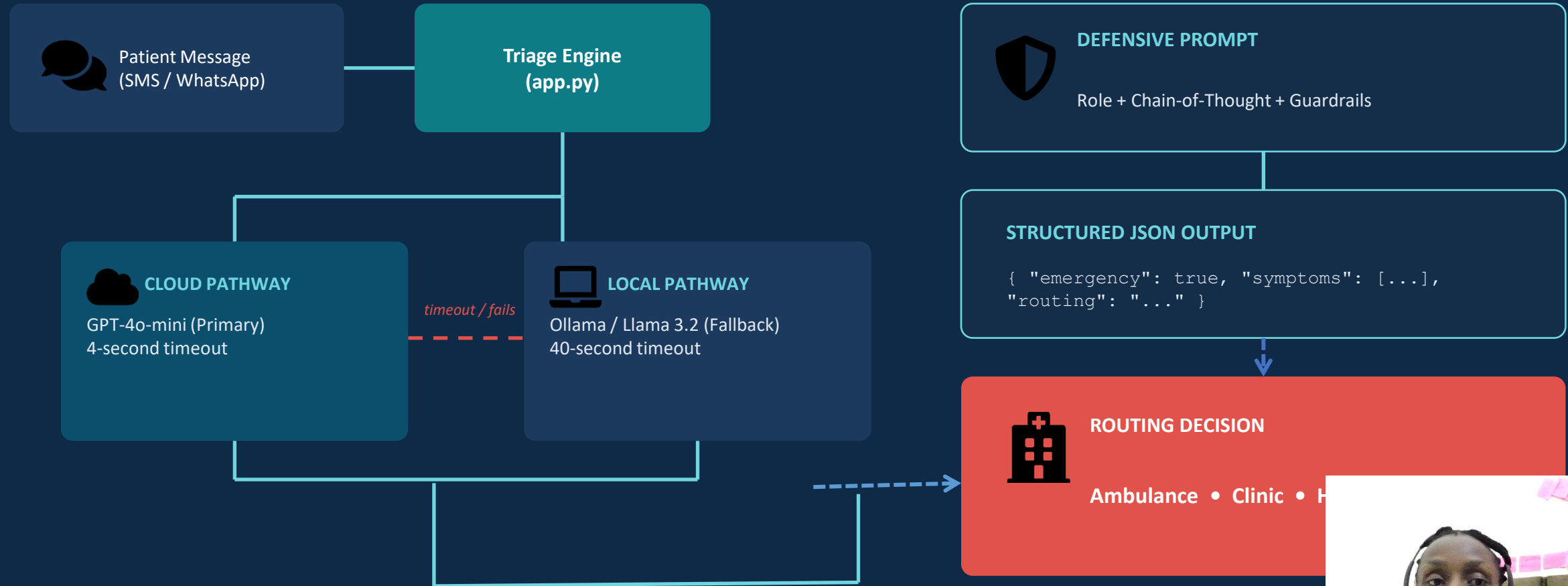


Result: a defensive gateway that fails safe, not silent — every response is validated before it ever reaches a patient.

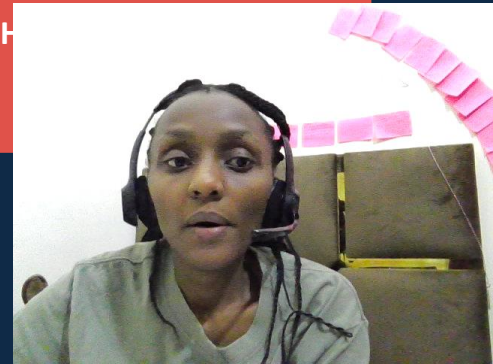


THE ARCHITECTURE

A Two-Track System for Zero Failure

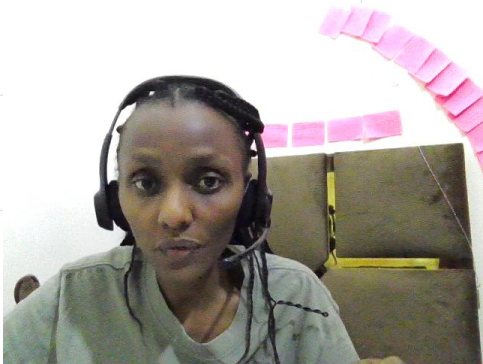


Every response passes through validation before delivery — the fallback guarantees zero silent failures.



What We've Fixed

Issue	Before (v1 / v2)	After (v3)
Diagnosis	"You have malaria"	"I cannot diagnose"
Prescriptions	"Take 400mg ibuprofen"	"Consult a doctor"
Speculation	"This could be cancer"	Symptoms only
Fluff	"I understand how you feel"	Direct, clinical
Markdown	```json fences	Raw JSON only
Network	System crashes	Auto-fallback
Parsing	Fails 20% of the time	100% valid JSON



THE NUMBERS

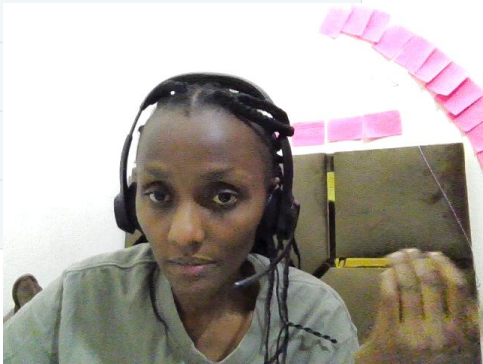
What This Means for the Business

Cost Savings per 10,000 Consultations

Cost Center	Without Guardrails	With Guardrails	Savings
API Tokens	\$380	\$152	\$228 (60% less)
Legal Risk	\$50,000+	\$0	\$50K+ avoided
Support Tickets	450 tickets	90 tickets	80% fewer
System Downtime	30% failure	0% failure	100% uptime

What We Can Now Scale

Metric	Before	After	Growth
Daily Patients	500	10,000+	20x
Geographic Reach	1 city	Entire country	Unlimited
Languages	1	5+	5x
Channels	Web only	WhatsApp + SMS + USSD	4x
Partners	None	Pharmacies + Government	New revenue





Safe by Design, Scalable by Architecture

AfyaPlus v3 turns unpredictable AI into a defensible, production-ready health triage system — cutting cost, closing legal risk, and reaching every patient, in every language, on every channel.

